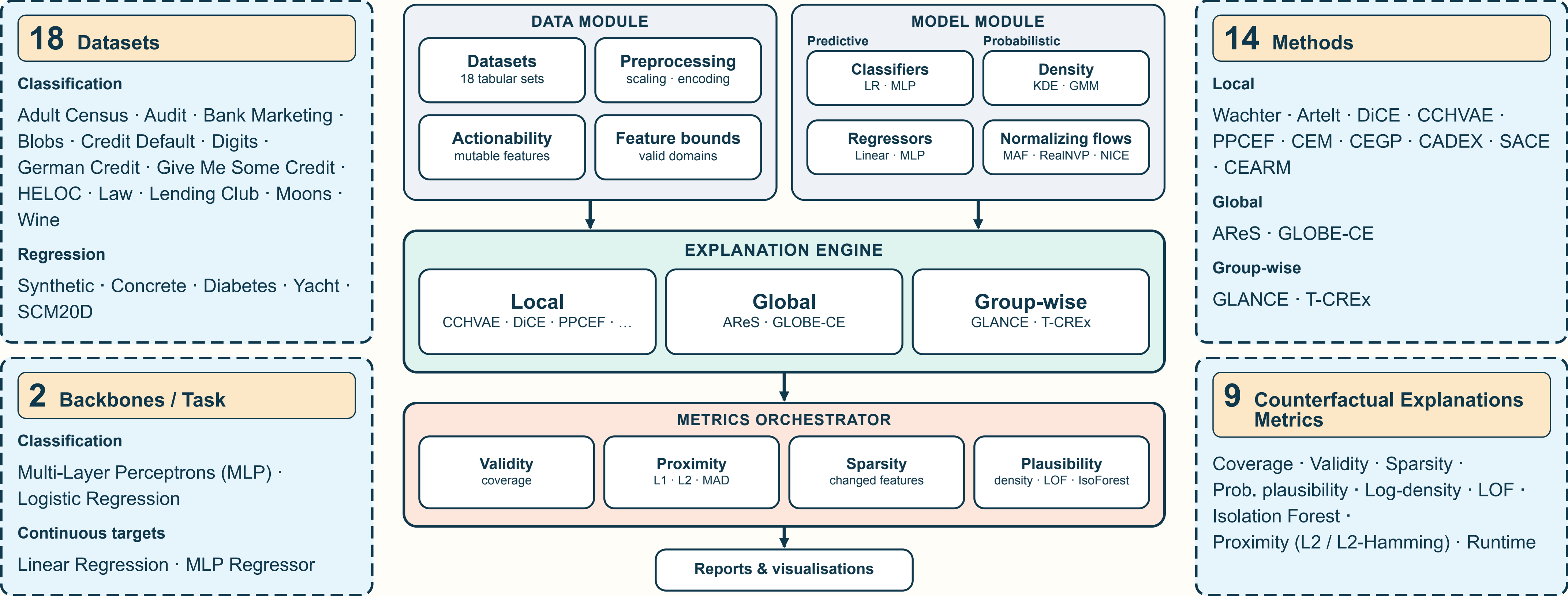




14 Methods

Local
Wachter · Artelt · DiCE · CCHVAE · PPCEF · CEM · CEGP · CADEX · SACE · CEARM
Global
ARes · GLOBE-CE
Group-wise
GLANCE · T-CREx

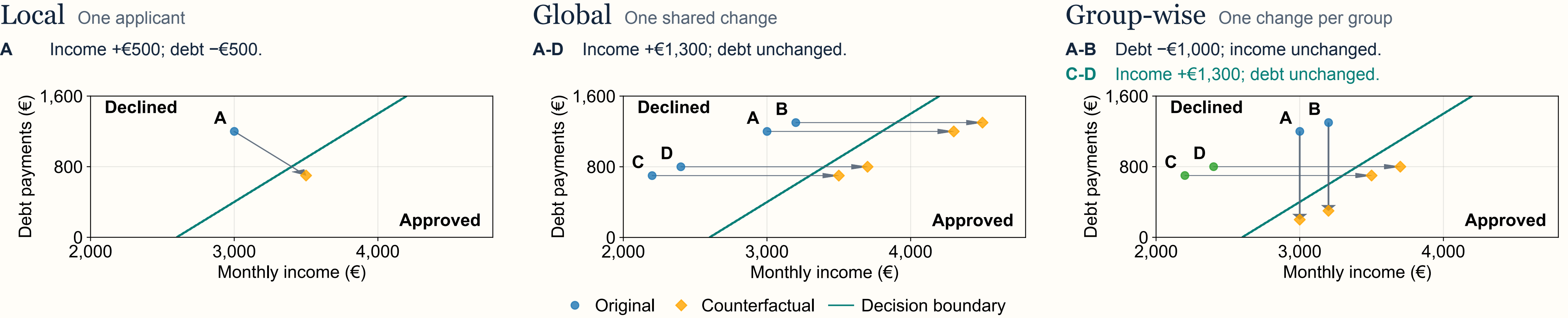
9 Counterfactual Explanations Metrics

Coverage · Validity · Sparsity · Prob. plausibility · Log-density · LOF · Isolation Forest · Proximity (L2 / L2-Hamming) · Runtime

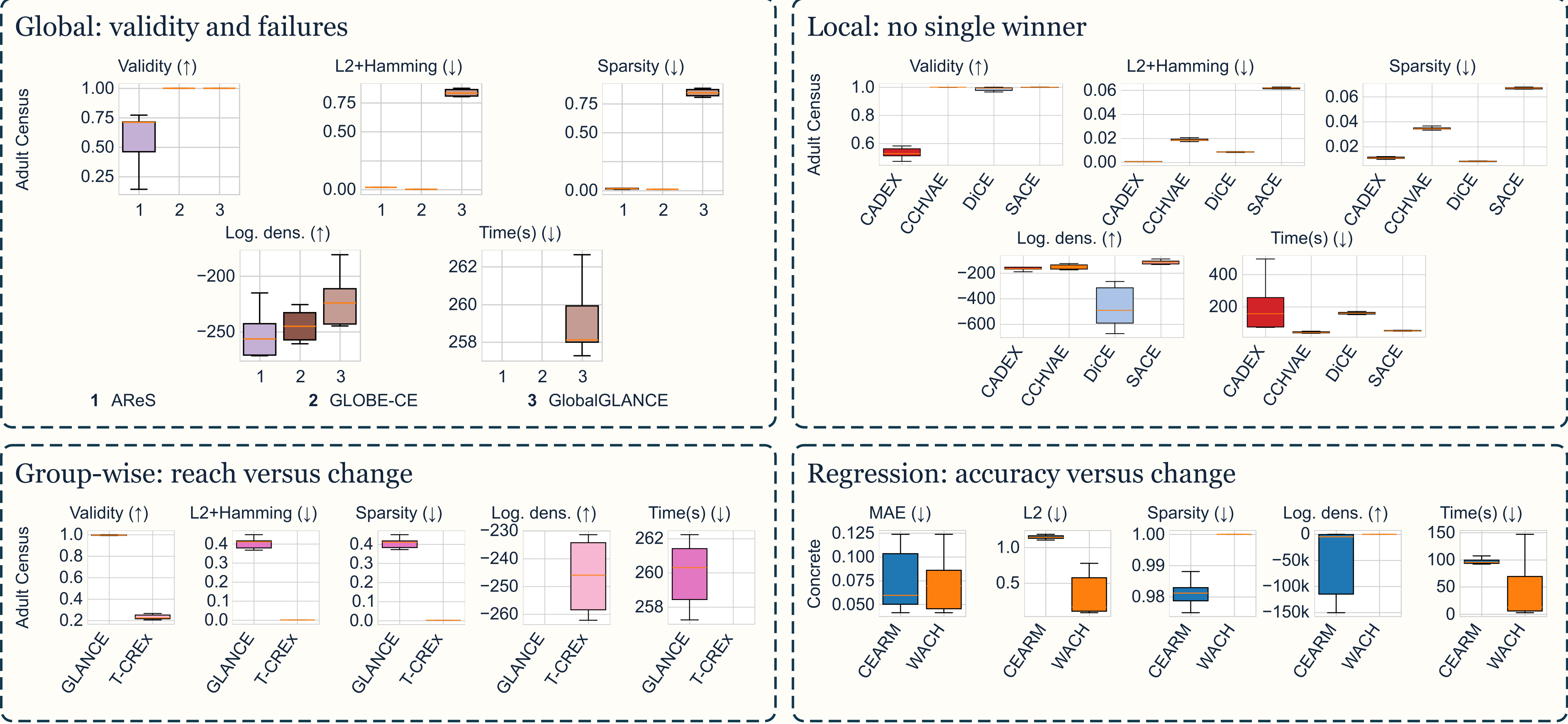
What is a counterfactual explanation?

A **counterfactual explanation** describes the smallest, realistic change in input data needed to flip an algorithmic decision.

It answers the “what-if” question by showing how a situation would look if specific facts were different. For example: “If your annual income was \$500 higher, your loan would have been approved.”



Results



Contributions

- 01 **Controlled protocol**
Reproducible evaluation
- 02 **Cross-paradigm benchmark**
Local · Global · Group-wise
- 03 **Extensible library**
Open source

